

C964: Computer Science Capstone

Task 2 parts A, B, C and D

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Part A: Letter of Transmittal

11/29/2025

Ramon S. Kist

SummitPoint Capital

1002 W Pleasant St

Knoxville, Iowa, 50138

Dear Mr. Kist,

Enclosed is the proposal titled “**SummitPoint Capital Credit Card Fraud Detection System Using Machine Learning**”. This proposal outlines a comprehensive plan to design, develop, and deploy an intelligent fraud detection system capable of identifying and preventing fraudulent credit card transactions before they happen.

SummitPoint Capital recently dealt with a spike in credit card fraud and has incurred severe financial losses as a result, as well as a loss in reputation amongst the regional financial industry. These losses are likely to increase as cyber criminals develop new and innovative ways to commit credit card fraud. This growing problem already threatens the growth of SummitPoint Capital and must be addressed; otherwise, we risk losing customers and losing market share to our industry rivals.

My Team and I propose a solution to create a machine learning-based detection system that will use existing fraud datasets to train our model. Logistic regression is the supervised machine learning algorithm that will be used to train our model, since it is a great algorithm for binary classification tasks such as fraud detection (fraud vs not-fraud). The model's effectiveness then will be determined by measuring its precision, accuracy, and recall. Hyperparameters will later be tuned to ensure we reach our desired precision, accuracy, and recall, which will be based on how much risk mitigation we desire vs inconveniencing customers for false positive detections. Finally, we will develop an application that can detect credit card fraud based on the parameters we have selected.

This fraud detection system can help detect credit fraud transactions. This will greatly reduce the risk to SummitPoint Capital and reduce any financial losses or compensation they

would have to pay out. This solution would also greatly improve customer satisfaction in two ways. Customer satisfaction has been found to increase when they see a bank proactively protecting them from fraud (Vieras et al., 2025). The second way is that it would protect SummitPoint Capital's reputation, which is significantly linked to customer satisfaction/dissatisfaction (Vieras et al., 2025).

This project will take 30 days (1month) to complete over 5 phases. These phases are data collection and analysis, model training, testing, application development, and deployment. The estimated cost of this project is \$16,700, which includes hardware, software, and human resource costs. Since we are handling credit card data, we will follow PCI DSS standards for the processes, storage, or transmission of any credit card information. We will ensure SOX compliance with all our systems to ensure information security, data integrity, and prevent fraud. Finally, we will adhere to GDPR and CCPA privacy laws to ensure user data is protected and to ensure customer satisfaction.

I fully believe that this project proposal is worth pursuing, and with my background in data science and information security, this project has a high chance of success and will yield great benefits for rather small costs. This project is based on proven methods for credit card fraud detection, and preventing fraud has been shown to increase customer satisfaction and company reputation. Finally, I respectfully request that you review this proposal and give approval to proceed with this project. Thank you for your time and consideration.

Sincerely,

Cole Kreiss

Cole Kreiss, Machine Learning Engineer

Part B: Project Proposal Plan

Project Summary

SummitPoint Capital has seen a recent rise in credit card fraud. This problem has been negatively affecting customer satisfaction and SummitPoint Capital's reputation and has also led to financial losses for SummitPoint Capital. These issues will compromise the long-term growth of SummitPoint Capital if we do not find a way to address this increase in fraud activity. This is because customer satisfaction and a bank's reputation are critical drivers of bank growth. If SummitPoint Capital wishes to continue growing its growth, market share, and profit, it will have to innovate and modernize its existing fraud detection systems. This is why, to address this problem, I propose that, on top of our legacy systems, we also use machine learning to detect credit card fraud.

SummitPoint Capital's needs are due to the existing inadequacy of its current fraud detection systems to detect new ways that cybercriminals are committing fraud. This can be addressed by using Machine learning to improve our fraud detection in real time. Machine learning can be used to improve fraud detection by detecting anomalous credit card activity that is unusual for a customer through analyzing prior behavior, and by identifying patterns that can be integrated into our existing rule-based legacy systems. By identifying these patterns and anomalies, we will improve our ability to detect fraud, and we can ensure customer satisfaction and SummitPoint Capital's brand reputation, which has recently declined and poses a risk to our long-term growth.

The deliverables for this project will be a machine learning model that can detect credit card fraud by using several independent variables for prediction. A web application prototype and a user guide that can be used to predict if a credit card transaction is fraudulent or not based on the filled-out parameters. Plans to integrate our model into our credit card processing infrastructure and legacy fraud prevention systems for real-time fraud analysis, prediction, and prevention.

The model that will be developed will improve our legacy fraud detection systems by detecting anomalous behavior and identifying patterns that it currently can't recognize. Our prototype web application will show how our model can be effective in fraud prediction and how it will work when integrated into our existing credit card transaction infrastructure in real-time. The ability to identify anomalous user activity and new cybercriminal patterns in real-time will help vastly improve our fraud detection capabilities. This proposal for a fraud detection system using machine learning will resolve SummitPoint Capital's immediate needs to reduce credit card fraud by improving our fraud detection. This reduction in fraud will overturn the recent loss

in customer satisfaction and SummitPoint Capital's reputation. Both of which are vital for the long-term growth of the bank.

Data Summary

The dataset we will be using for this project is a "Credit Card Fraud" dataset from Kaggle (Narayanan, 2022). This data will be downloaded as a .csv file and then will be accessed by importing pandas, which will load it into a data frame so we can explore and modify the data using Python.

Throughout the application development life cycle, the data we use will be carefully managed to ensure security and compliance standards are met on workstations that are handling the data, and identify immediately if the dataset has sensitive information. In the exploration phase, the dataset will then be inspected to understand its structure, size, and distribution, and visualizations will be generated to assist in understanding the data. Next, the Data will be cleaned in the modify phase if any duplicates, missing values, errors, or outliers are found that could negatively affect our model. In the modeling phase, data will be normalized, scaled, and split into training, validation, and test sets. In the Assess phase, data will be assessed for performance, and hyperparameters will be tuned. Only our model and the processed data will be used in the application. The application will then be monitored and maintained to identify if our dataset was useful for prediction. Finally, this dataset will be archived, and documentation will be created on how this data was used and processed.

The dataset we are using contains data columns for `distance_from_home`, `distance_from_last_transaction`, `ratio_to_median_purchase_price`, `repeat_retailer`, `used_chip`, `used_pin_number`, and `online_order`. These are data columns that we can create and track from our own real production data so that we can perform real-time fraud detection when credit card transactions occur. Incomplete data will be replaced with predicted values using regression imputation. Standardized residuals will be calculated and used to detect anomalies and outliers. These will be assessed to determine if they are legitimate data points or errors. If the data points are legitimate, they will be used, and if they are considered to be errors, they will be removed.

The data from Kaggle is considered public by the publisher, so no licensing is required for its use. We will follow information security best practices and compliance standards to ensure no exploitation can occur from this project and to comply with Summit Capital's legal and ethical obligations. Standards we will abide by are SOX and PCI-DSS. This would require us to conduct risk assessments. We must identify data, encrypt data, implement access controls to our development environment, secure our environment, and train our employees. Implementing these standards now will ensure we remain compliant and maintain our ethical standards now and in the future.

Implementation

We will use the SEMMA methodology to develop our credit card fraud detection system. Our fraud detection model uses a Kaggle dataset as our sample that we will explore and modify. From there, we will develop our model and assess its effectiveness.

- Sample:

For our initial dataset, we will be using the Kaggle Credit Card Fraud Dataset Suite (Narayanan, 2022). The dependent variable is the Boolean fraud variable, i.e., 1 if fraud, 0 if not. The dependent variables that our model will use for prediction are `distance_from_home`, `distance_from_last_transaction`, `ratio_to_median_purchase_price`, `repeat_retailer`, `used_chip`, `used_pin_number`, and `online_order`. This dataset will be split for 90% Training, 5% Validation, and 5% Testing since the dataset is very large with 1,000,000 data points.

- Explore:

In this phase, we will derive the mean, median, standard deviation, and skewness for independent variables using histograms. Find the correlations between the dependent fraud variable and independent variables to determine which variables are the strongest predictors using a correlation matrix. Assess outliers within the data and determine if they are valuable or disposable. Develop several preliminary hypotheses about what variables most likely contribute to fraud.

- Modify:

In this phase, we will clean the dataset for any duplicates and outliers that we determined were unnecessary. Fill in missing values using imputation. Remove any variables that do not correlate with bank account fraud detection. Determine which variables are most important and prioritize them, as this will be used to improve our model and the magnitude of our coefficients when using logistic regression.

- Model:

In this phase, we will determine the model technique to use, and in our case, this will be multiple logistic regression models. We will then divide the dataset into training, validation, and testing. Create different parameters and coefficients for each model based on our hypotheses.

- Assess:

In the Assess phase, we will evaluate our final model performance by testing it on our validation data. We will then gather common metrics for each regression model, such as accuracy, precision, recall, and F-1-score. Next is to select the model that balances accuracy, interpretability, and simplicity. Finally, we'll evaluate the cost-benefits, summarize our results, and determine if the model is satisfactory enough to move into building our application.

Timeline

Milestone or deliverable	Project Dependencies	Resources	Start and End Date	Duration
• Acceptance of Proposal	N/A	Machine Learning Engineer	01/01/26-01/02/26	1 days
• Workstation Preparation • Set up workstation for model development. • Ensure our workstation is SOX compliant.	Acceptance of Proposal	Machine Learning Engineer Information Security Engineer Endpoint Engineer Workstation	01/02/26-01/03/26	1 days
• Sample Phase • Gather fraud bank account creation data	Workstation Preparation	Machine Learning Engineer Workstation Kaggle Dataset	01/03/26-01/04/26	1 days

suite that will be used to train our model. Classify the data and perform anonymization.				
- Explore Phase - Perform descriptive statistics on independent variables. - Identify missing values, outliers, and anomalies. - Explore differences between fraudulent vs legitimate transactions.	Sample Phase	Machine Learning Engineer Workstation Kaggle Dataset	01/04/26-01/07/26	3 days
- Modify Phase - Handle missing data and outliers in SummitPoint Capital data. - Normalize/standardize independent data columns and balance the dataset	Explore Phase	Machine Learning Engineer Workstation Kaggle Dataset	01/07/26-01/12/26	5 days
- Model Phase - Split data into training, test, and validation sets.	Modify Phase	Machine Learning Engineer Workstation Kaggle Dataset	01/12/26-01/19/26	7 days

• Train the logistic regression model using our training set. • Use GridSearchCV for cross-validation and tuning parameters • Evaluate interim performance using metrics like accuracy, precision, recall, and F1-score				
• Assess Phase • Evaluate our model using our test data. • Perform robustness checks across time and customer segments. • Calculate logistic regression standardized residuals and check for outliers that weaken our model.	Model Phase	Machine Learning Engineer Workstation Kaggle Dataset	01/19/26-01/22/26	3 days
• Prototype Development	Assess Phase	Machine Learning Engineer	01/22/26-01/27/26	5 days

• Create a web application prototype that uses our model to make a prediction on whether a credit card transaction is fraud based off the entered parameters • Test our prototype by using real-world credit card fraud data from our fraud analysis team.		Workstation Model Pipeline		
• Submit the Prototype for Review • Gather senior leadership's opinion on the prototype and ask to move forward with drafting a proposal to integrate our model into our credit card transaction infrastructure	Prototype Development	Machine Learning Engineer Workstation Prototype	01/27/26- 01/29/26	2 days
• Draft plan for deploying the fraud detection model in our credit card	Prototype is accepted or rejected	Machine Learning Engineer Workstation Prototype	01/29/26- 01/31/26	2 days

transaction infrastructure for real-time analysis Documentation will be created on how we created our model, prototype, and our performance metrics		Real-time detection proposal plan		
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Evaluation Plan

The visualized verification methods used in this project are histograms, correlation matrix heatmaps, and performance metric dashboards. Histograms will help reveal the distribution of our fraud and non-fraud datapoints for each independent variable. Correlation matrix heatmaps will show if any multicollinearity exist between variables and how related each variable is to predict fraud. Finally, our performance metric dashboards will show accuracy, precision, recall, and F-1 scores. These dashboards help us visualize how our model performs at prediction on our test set.

N-fold cross validation is the validation method we are using for this project. Our Kaggle dataset will be split into 3 sets. 90% of the datapoints will go to training, 5% of our dataset will be used for validation, and 5% of our data will be used for testing. Most of our datasets will be used for training since we have a dataset that is very large (1,000,000 datapoints).

Costs

Resource	Description	Estimated Cost
Hardware	Workstation (laptop)	\$1,500
Data	Kaggle dataset is public and requires no licensing	\$0

Human Resources	Machine Learning Engineer (30 days at \$60/hr for 8 hr days)	\$15,200
	Information Security Engineer (1 day at \$50/hr for 8 hr days)	
	Endpoint Engineer (1 day at \$50/hr for 8 hr days)	
Software	Python, Jupyter Notebook, Python Libraries (pandas, NumPy, matplotlib, seaborn, sklearn, streamlit)	\$0 (open source)
Total Cost:	Total cost of project	\$16,700

Part C: Application

Required Files:

- Credit Card Fraud Detection App.py - File used to generate the web application.
- Credit Card Fraud Detection Pipeline.pkl - This contains our model and ability to predict fraud.
- card_trans_fraud_data.csv - The Kaggle dataset used to train our model.
- Fraud Analysis Model.ipynb - The jupyter notebook used for data analysis and model development.

Part D: Post-implementation Report

Solution Summary

SummitPoint Capital had seen a recent rise in credit card fraud. This problem had negatively affected customer satisfaction, SummitPoint Capital's reputation, and led to financial losses for SummitPoint Capital. The proposed solution for a fraud detection system using machine learning helped resolve SummitPoint Capital's immediate needs to reduce credit card fraud by improving our fraud detection. This reduction in fraud overturned the recent loss in customer satisfaction and SummitPoint Capital's reputation. Both of which are vital for the long-term growth of the bank.

Data Summary

The dataset we used for this project was the "Credit Card Fraud" dataset from Kaggle (Narayanan, 2022). This data was downloaded as a .csv file and then accessed by importing pandas, which was used to load the dataset into a data frame so we could explore and modify the data using Python.

After our data was loaded into a data frame, basic information was gathered, such as mean, median, min values, and maximum values. Several visualizations were generated to understand the shape of our data and to form some preliminary hypotheses about what strategy fraudsters usually use to commit fraud. Descriptions of our data visualizations are posted below; however, an example of fraud patterns can be seen by looking at the contingency tables and how fraud is more likely to be committed online, without using an EVM chip, and having no PIN used. We also used these visualizations and min and max values to determine if any data points were errors and if any needed to be removed, such as a data point having a distance that is larger than half the circumference of the earth, which would be the maximum value for the distance variables.

After we performed our initial analysis, our data was split into 90% training, 5% validation, and 5% testing sets. 2 models were initially fit using our training dataset to determine how different a model would perform when balancing our training dataset and not balancing our training dataset. The differences were rather stark when comparing the precision and recall of the 2 different models. Higher precision and lower recall were observed for the model with no balancing, and lower precision and higher recall were observed for the balanced model. We attempted to balance recall and precision and arrived at a class_weight balance of 1:2.5. Further

hyperparameter optimization was performed using GridSearchCV. We inevitably found the best model to build our application.

The application that we built used the .pkl file that was generated from the best model we selected after hyperparameter tuning was complete. This .pkl file was used for the prediction aspect of our application.

Machine Learning

The machine learning method used to develop our fraud detection model was the logistic regression statistical method.

Logistic regression was used in the model to assess the SEMMA phases. Logistic regression uses a logistic function to transform the output of a linear equation into a probability value between 0 and 1, making it suitable for binary outcomes and, in our case, fraud (1) or not-fraud (0). Our model was developed by initially analyzing our data to remove errors, ensuring it fits the distribution of our dataset. No errors were observed, and 3 of our datapoints were left skewed, which is acceptable when using logistic regression for modeling. We then trained our model using a couple of different hyperparameters and then followed up using GridSearchCV to further improve our fraud detection model and to perform cross-validation. After tuning the model using the training and validation datasets, it was finally tested against our test set and achieved 96% accuracy, ~80% precision, and ~80% recall. Our model was then dumped into a .pkl file so it could be used for prediction in our web application.

Logistic regression is great for its simplicity in how it can efficiently compute, which is useful for a credit card transaction environment that is very fast, and how it can serve as an initial baseline for our credit card fraud detection system (Gupta et al., 2025). It was also used since logistic regression achieved similar effectiveness against different AI learning models for an AI bank account fraud detection system (Islam et al., 2025). This research article concludes that supervised learning models are effective given labeled fraud datasets, which are what our fraud analysts produce, and can help train and verify our model to meet our needs of reducing our workload. In summary, we selected logistic regression for its simplicity, effectiveness, and usefulness as a baseline model.

Validation

Logistic regression is the machine learning model that we used, which is a supervised machine learning method. Logistic regression is a supervised machine learning method since it uses labeled data sets to predict an outcome and, in our case, predict fraud.

We used GridSearchCV and specified that 2-fold cross-validation would be performed to find the most effective hyperparameters. Performance metrics were generated when performing grid search, which was averaged from the 2 subsets of data we specified by using 2-fold-cross-validation. We prioritized f-1 scoring and monitored accuracy, precision, and recall when running GridSearchCV. Monitoring accuracy, precision, recall will allow us to create filters and pick the model with the highest f-1 score remaining.

A classification report was used to validate the model's performance. A classification report provides the accuracy, precision, recall, and F-1score of our model. We tuned the weight_balance hyperparameter to have ensure a balance between precision and recall. F-1 score was the prioritized metric since F-1 score will allow us to detect and block as many of the actual positive fraud cases as possible without blocking too many legitimate transactions that could irritate our customers. We achieved a model with 96% accuracy, ~80% precision, and ~80% recall which is great for a first model.

Classification Report:

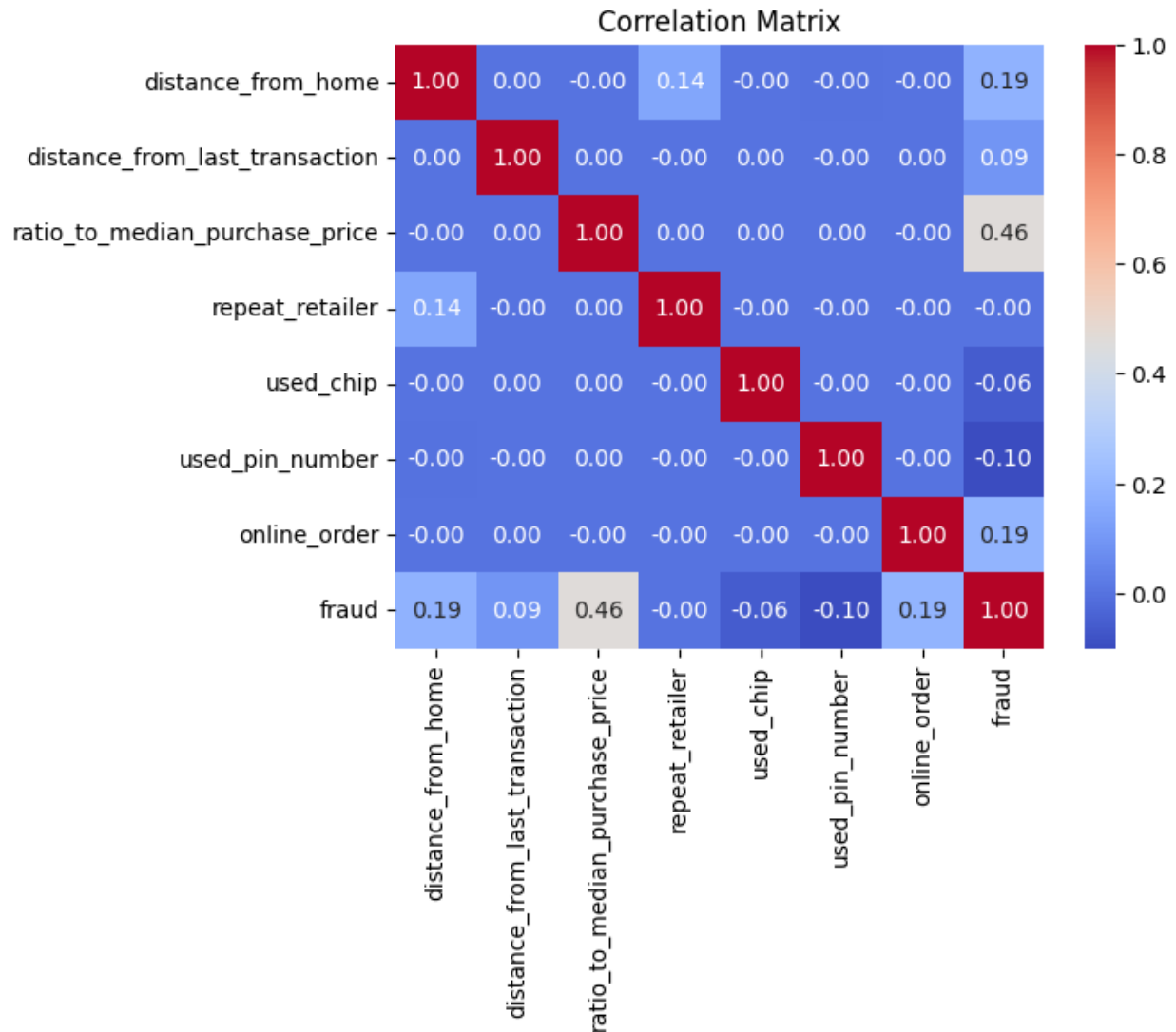
-	Precision	Recall	F1-Score	Support
0.0 (Not-Fraud)	.98	.98	.98	45689

1.0 (Fraud)	.79	.79	.79	4311
Accuracy	-	-	.96	50000
Macro avg	.89	.89	.89	50000
Weighted avg	.96	.96	.96	50000

Visualizations

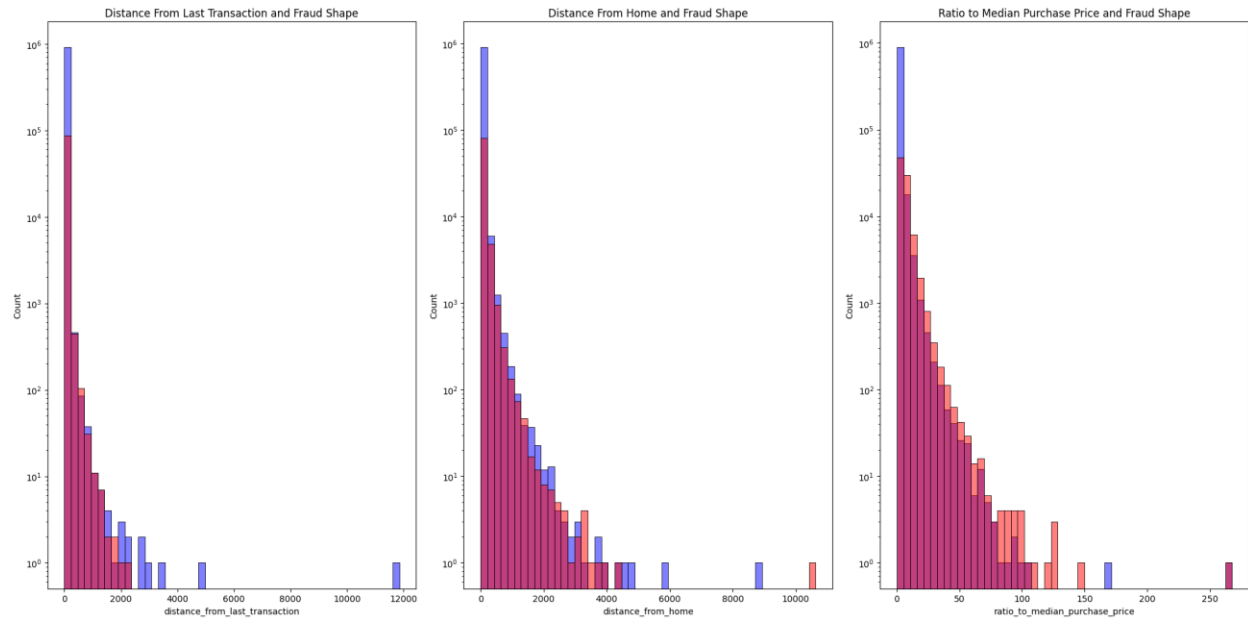
Correlation Matrix:

This correlation matrix shows the correlation between independent and dependent variables that our model uses to detect fraud. We can see that there is low multicollinearity; however, there is a low correlation between `distance_from_home` and `repeat_retailer`. Low multicollinearity between variables is desirable for a logistic regression-trained model, which we are using.



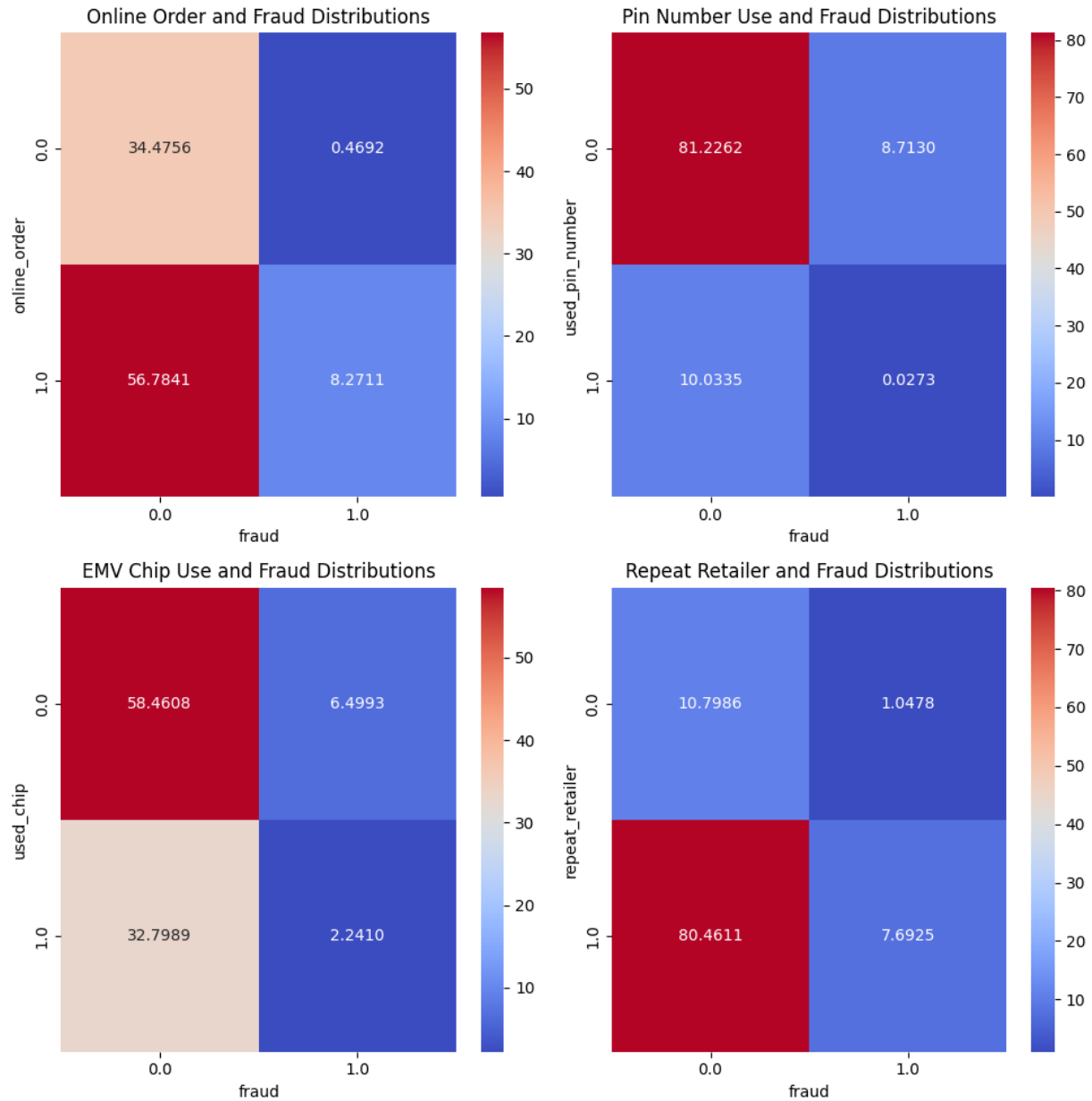
Histogram:

These 3 histograms show the distributions and skew of the 3 independent variables `distance_from_home`, `distance_from_last_transaction`, and `ratio_to_median_purchase_price` for both fraud (red colored bars) and non-fraud (blue colored bars) transactions. Below, you can see that for both fraud and non-fraud transactions, the 3 dependent variables are left-skewed. However, there are still differences between the fraud and non-fraud distributions, which indicate these variables are valuable for prediction, and logistic regression does not assume a normal distribution; thus, it will serve as a useful model for training when using this dataset.



Contingency Table:

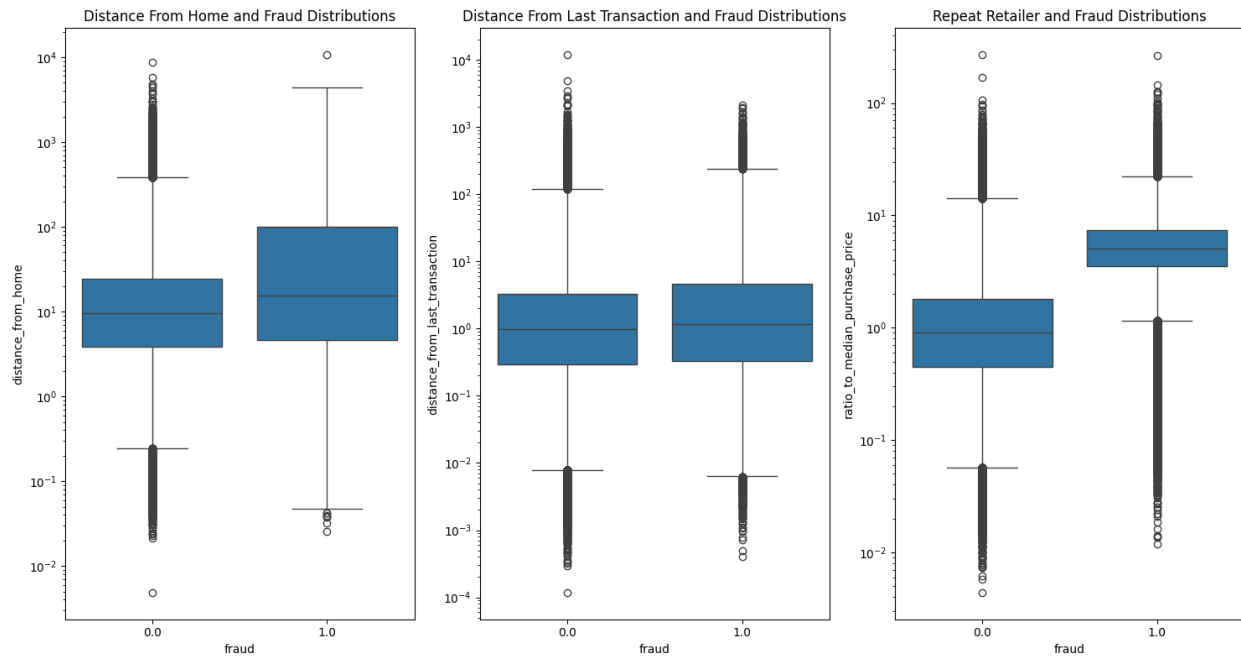
Below are 4 contingency tables which show the distributions of our 4 dependent binary variables by fraud vs non-fraud. The distributions reveal the relationship between fraud and the dependent variable. For example, the `online_order` contingency table has a rather even split between online orders and offline orders when there is no fraud. However, when there is fraud, it is much more likely that the order was an online order, so we can predict that a decent-sized coefficient will be used for the `online_order` parameter when our model is fully trained.



Box Plots:

The 3 box plots below show us the distribution of the 3 variables `distance_from_home`, `distance_from_last_transaction`, and `ratio_to_median_purchase_price` based on fraud. The usefulness of these boxplots is that they visualize how many outliers there are and reveal any significant outliers. These figures show us several interesting data points that we investigated to determine if they should be removed. An example would be for the 2 outliers for the “Distance From Home and Fraud Distributions” box blots. One data point shows us that the distance from home was almost zero, which is not unreasonable. The other was a data point of 11,851 miles

from home and is below the possible maximum distance from home of 12,500 miles, which is half the circumference of the Earth.



Downloading the dataset and Jupyter notebook files for full use:

1. Install VScode and Python. Via the command line install jupyter and ipykernel. We'll be using the jupyter kernel Python 3.13.9. Install via the command line seaborn, pandas, scikit-learn, matplotlib, numpy, streamlit, joblib, and pandas if not already installed.
2. Download the files "Fraud_Analysis_Model.ipynb" and "Credit_Card_Fraud_Detection_App.py" from [Github](#) to your vscode folder.
3. Download from Kaggle the .csv file dataset from [Credit Card Fraud Dataset](#) to your vscode folder. Rename the csv file to "card_trans_fraud_data.csv"
4. Make sure you're using Jupyter Kernel "Python 3.13.9". Switch to our jupyter notebook file and click the "Run All" button in vscode. This should generate a .pkl file which we will use for our predictions.
5. Run the command "streamlit run Credit_Card_Fraud_Detection_App.py".
6. Connect to the webapp via your browser by using the Local URL: provided by the above command.
7. Fill out the fields and click the "Predict" button to generate a credit card fraud prediction.

Downloading Just the webapp and model files:

1. Install vscode and python. Via the command line install streamlit, joblib, and pandas if not already installed.
2. Download the files "Credit_Card_Fraud_Detection_App.py" and "Credit_Card_Fraud_Detection_Pipeline.pkl" from [Github](#)
3. Run the command "streamlit run Credit_Card_Fraud_Detection_App.py".
4. Connect to the webapp via your browser by using the Local URL: provided by the above command.
5. Fill out the fields and click the "Predict" button to generate a credit card fraud prediction.

Reference Page

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